DOKUMEN PANDUAN TEKNIS & TUTORIAL

Tutorial Lengkap Pembuatan Aplikasi Sistem Informasi Toko Obat (Apotek)

Panduan langkah demi langkah implementasi kode Windows Forms C# .NET 8 dan Microsoft SQL Server dengan arsitektur multi-role: Halaman Login → Admin → Kasir → Owner.

BAHASA PEMROGRAMAN	DATABASE ENGINE	FORMAT PENYUSUNAN
C# 12 / .NET 8.0 WinForms	Microsoft SQL Server (T-SQL)	Penjelasan Fungsi per Fungsi

FONDASI SISTEM

Bab 0: Fondasi Database & Kelas Inti

Sebelum masuk ke tampilan halaman, aplikasi butuh koneksi database SQL Server dan sistem untuk mengingat siapa pengguna yang sedang login.

Struktur Database SQL Server (tokoobat.sql)

Skema Tabel Utama Apotek

DATABASE SCHEMA

BUAT APA FUNGSI INI?

Mendefinisikan 6 tabel utama: Users (akun pegawai & role), Produk (stok & harga obat), Pelanggan (data customer), Transaksi (header penjualan), DetailTransaksi (item obat yang terjual), dan LogAktivitas (rekam jejak seluruh aksi).

tokoobat.sql — Skema Tabel Utama Apotek

```
-- Tabel Akun Pengguna (Admin, Kasir, Owner)
CREATE TABLE Users (
    Id INT IDENTITY(1,1) PRIMARY KEY,
    Nama VARCHAR(100) NOT NULL UNIQUE,
    Password VARCHAR(255) NOT NULL,
    Role VARCHAR(50) NOT NULL,
    FotoProfil VARBINARY(MAX) NULL
);

-- Tabel Data Obat / Produk
CREATE TABLE Produk (
    Id INT IDENTITY(1,1) PRIMARY KEY,
    IdAdmin INT NOT NULL,
    Nama VARCHAR(100) NOT NULL,
    Category VARCHAR(100) NOT NULL,
    TanggalKadaluarsa DATE NOT NULL,
    Stok INT NOT NULL,
    Jenis VARCHAR(50) NOT NULL,
    Harga VARCHAR(50) NOT NULL,
    FotoProduk VARBINARY(MAX) NULL,
    Deskripsi VARCHAR(255) NULL
);

-- Tabel Transaksi Kasir
CREATE TABLE Transaksi (
    Id INT IDENTITY(1,1) PRIMARY KEY,
    IdKasir INT NOT NULL,
    IdPelanggan INT NULL,
    TotalProduk INT NOT NULL,
    Harga_Total INT NOT NULL,
    Uang_Bayar INT NOT NULL,
    Uang_Kembali INT NOT NULL,
    Tanggal_Transaksi DATETIME DEFAULT GETDATE()
);
```

Kelas Sentral (Koneksi & Sesi) (TokoObatClasses.cs)

Databases.GetConn()

METHOD

BUAT APA FUNGSI INI?

Fungsi ini dipakai untuk membuka jembatan koneksi ke database SQL Server lokal dengan TrustServerCertificate=True.

TokoObatClasses.cs — Databases.GetConn()

```
class="tok-kw">public class="tok-kw">static SqlConnection GetConn()
{
    class="tok-kw">string connString = @class="tok-class="tok-kw">string">"Data
Source=localhost;Initial Catalog=tokoobat;Integrated Security=True;TrustServerCertificate=True;";
    class="tok-kw">return class="tok-kw">new SqlConnection(connString);
}
```

Databases.CatatLog(string pesanAktivitas)

METHOD

BUAT APA FUNGSI INI?

Fungsi otomatis buat mencatat setiap aksi pegawai (seperti login, tambah obat, transaksi) ke dalam tabel LogAktivitas biar Owner bisa audit.

TokoObatClasses.cs — Databases.CatatLog(string pesanAktivitas)

```
class="tok-kw">public class="tok-kw">static class="tok-kw">void CatatLog(class="tok-kw">string
pesanAktivitas)
{
    class="tok-kw">try
    {
        class="tok-kw">using (SqlConnection conn = GetConn())
        {
            conn.Open();
            class="tok-kw">string query = class="tok-class="tok-kw">string">"INSERT INTO
LogAktivitas (IdUser, Aktivitas, Waktu) VALUES (@idUser, @pesan, GETDATE())";
            class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
            {
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@idUser",
UserSession.Id > 0 ? UserSession.Id : (object)DBNull.Value);
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@pesan",
pesanAktivitas);
                cmd.ExecuteNonQuery();
            }
        }
    }
    class="tok-kw">catch { /* Jangan sampai log menggagalkan proses utama */ }
}
```

UserSession.SetUser(...) & LakukanLogout(...)

METHOD

BUAT APA FUNGSI INI?

Fungsi SetUser menyimpan ID, Username, dan Role siapa yang login. Fungsi LakukanLogout membersihkan sesi, mencatat aktivitas logout, lalu mengembalikan layar ke Form Login.

TokoObatClasses.cs — UserSession.SetUser(...) & LakukanLogout(...)

```
class="tok-kw">public class="tok-kw">static class="tok-kw">void SetUser(class="tok-kw">int id,
class="tok-kw">string username, class="tok-kw">string role)
{
    Id = id;
    Username = username;
    Role = role;
}

class="tok-kw">public class="tok-kw">static class="tok-kw">void LakukanLogout(Form formAktif)
{
    DialogResult konfirmasi = MessageBox.Show(
        class="tok-class="tok-kw">string">"Apakah Anda yakin ingin logout dari sistem?",
        class="tok-class="tok-kw">string">"Konfirmasi Logout",
        MessageBoxButtons.YesNo,
        MessageBoxIcon.Question
    );

    class="tok-kw">if (konfirmasi == DialogResult.Yes)
    {
        Databases.CatatLog(class="tok-class="tok-kw">string">"Logout dari sistem");
        Clear();

        Form? loginForm = null;
        class="tok-kw">foreach (Form f class="tok-kw">in Application.OpenForms)
        {
            class="tok-kw">if (f is Halaman_Login)
            {
                loginForm = f;
                class="tok-kw">break;
            }
        }

        class="tok-kw">if (loginForm != null)
        {
            loginForm.Show();
            formAktif.Close();
        }
        class="tok-kw">else
        {
            class="tok-kw">new Halaman_Login().Show();
            formAktif.Close();
        }
    }
}
```

AUTENTIKASI

Bab 1: Halaman Login (Halaman_Login)

Gerbang utama sistem untuk menyaring siapa yang masuk, memastikan password benar, dan mengarahkan pengguna ke dashboard sesuai wewenangnya.

Form Login (Halaman_Login.cs)

`Halaman_Login() (Constructor)` CONSTRUCTOR

BUAT APA FUNGSI INI?

Dijalankan pertama kali saat aplikasi dibuka. Mengatur agar layar langsung Fullscreen (Maximized) dan jika jendela login ditutup maka seluruh aplikasi berhenti.

Halaman_Login.cs — Halaman_Login() (Constructor)

```
class="tok-kw">public Halaman_Login()
{
    InitializeComponent();
    this.WindowState = FormWindowState.Maximized; class="tok-comment">// Auto Fullscreen
    this.DoubleBuffered = true;
    this.FormClosed += (s, e) => Application.Exit();
}
```

`button1_Click(object sender, EventArgs e)`

EVENT HANDLER (TOMBOL LOGIN)

BUAT APA FUNGSI INI?

Mengecek apakah username dan password yang diketik ada di tabel Users. Jika valid, sesi login disimpan dan aplikasi membuka Dashboard yang sesuai: Admin, Kasir, atau Owner.

Halaman_Login.cs — button1_Click(object sender, EventArgs e)

```
class="tok-kw">private class="tok-kw">void button1_Click(object sender, EventArgs e)
{
    class="tok-kw">string username = txtUsername.Text.Trim();
    class="tok-kw">string password = txtPassword.Text.Trim();

    class="tok-kw">if (class="tok-kw">string.IsNullOrEmpty(username) || class="tok-
kw">string.IsNullOrEmpty(password))
    {
        lblError.Text = class="tok-class="tok-kw">string">"Harap isi username dan password.";
        class="tok-kw">return;
    }

    class="tok-kw">try
    {
        class="tok-kw">using (SqlConnection conn = Databases.GetConn())
        {
            conn.Open();
            class="tok-kw">string query = class="tok-class="tok-kw">string">"SELECT Id, Nama, Role
FROM Users WHERE Nama = @username AND Password = @password";
            class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
            {
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@username",
username);
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@password",
password);

                class="tok-kw">using (SqlDataReader reader = cmd.ExecuteReader())
                {
                    class="tok-kw">if (reader.Read())
                    {
                        class="tok-kw">int idUser = Convert.ToInt32(reader[class="tok-class="tok-
kw">string">"Id"]);
                        class="tok-kw">string roleAkun = reader[class="tok-class="tok-
kw">string">"Role"].ToString();

                        class="tok-comment">// Simpan sesi login saat ini
                        UserSession.SetUser(idUser, username, roleAkun);
                        Databases.CatatLog($class="tok-class="tok-kw">string">"Login ke sistem
sebagai {roleAkun}");

                        txtPassword.Clear();
                        lblError.Text = class="tok-class="tok-kw">string">"";

                        class="tok-comment">// Pindah ke halaman sesuai Role
                        class="tok-kw">if (roleAkun == class="tok-class="tok-kw">string">"Admin")
```

```
                        {
                            class="tok-kw">new Admin_Dashboard().Show();
                            this.Hide();
                        }
                        class="tok-kw">else class="tok-kw">if (roleAkun == class="tok-class="tok-
kw">string">"Kasir")
                        {
                            class="tok-kw">new Kasir_Dashboard().Show();
                            this.Hide();
                        }
                        class="tok-kw">else class="tok-kw">if (roleAkun == class="tok-class="tok-
kw">string">"Owner")
                        {
                            class="tok-kw">new Owner_Dashboard().Show();
                            this.Hide();
                        }
                    }
                    class="tok-kw">else
                    {
                        lblError.Text = class="tok-class="tok-kw">string">"Username atau password
salah!";
                    }
                }
            }
        }
    }
    class="tok-kw">catch (Exception ex)
    {
        MessageBox.Show(class="tok-class="tok-kw">string">"Gagal terhubung ke database: " +
ex.Message);
    }
}
```

MODUL ADMIN

Bab 2: Modul Admin (Pengelolaan Toko & Master Data)

Modul untuk administrator: memantau aktivitas sistem, mengelola akun seluruh pegawai, mengatur inventori obat lengkap dengan foto, dan mengedit profil pribadi.

Admin_Dashboard.cs (Admin_Dashboard.cs)

`MuatHalaman()`

METHOD

BUAT APA FUNGSI INI?

Mengambil ringkasan statistik (Total Transaksi, Total Akun Pegawai, Total Obat) dan menampilkan tabel 10 aktivitas pegawai terkini dari tabel LogAktivitas.

Admin_Dashboard.cs — MuatHalaman()

```
class="tok-kw">private class="tok-kw">void MuatHalaman()
{
    lblWelcome.Text = class="tok-class="tok-kw">string">"Selamat datang, " +
UserSession.Username.ToUpper();

    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();

        class="tok-comment">// Ambil 10 aktivitas terbaru di sistem
        class="tok-kw">string queryLog = class="tok-class="tok-kw">string">@"
            SELECT TOP 10
                l.Id AS [ID],
                COALESCE(u.Nama, 'Sistem') AS [Pengguna],
                l.Aktivitas AS [Aktivitas],
                l.Waktu AS [Waktu]
            FROM LogAktivitas l
            LEFT JOIN Users u ON l.IdUser = u.Id
            ORDER BY l.Waktu DESC";

        class="tok-kw">using (SqlDataAdapter da = class="tok-kw">new SqlDataAdapter(queryLog,
conn))
        {
            DataTable dt = class="tok-kw">new DataTable();
            da.Fill(dt);
            dgv10AktivitasTerkini.DataSource = dt;
        }

        class="tok-comment">// Hitung total angka untuk kartu metrik
        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(class="tok-class="tok-
kw">string">"SELECT COUNT(*) FROM Transaksi", conn))
            lblTotalTransaksi.Text = cmd.ExecuteScalar()?.ToString() ?? class="tok-class="tok-
kw">string">"0";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(class="tok-class="tok-
kw">string">"SELECT COUNT(*) FROM Users", conn))
            lblTotalAkun.Text = cmd.ExecuteScalar()?.ToString() ?? class="tok-class="tok-
kw">string">"0";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(class="tok-class="tok-
kw">string">"SELECT COUNT(*) FROM Produk", conn))
            lblTotalProduk.Text = cmd.ExecuteScalar()?.ToString() ?? class="tok-class="tok-
kw">string">"0";
```

```
    }
}
```

Navigasi Sidebar Admin

EVENT HANDLER (TOMBOL MENU)

BUAT APA FUNGSI INI?

Fungsi untuk berpindah antar halaman admin (Akun, Produk, Profil) dan tombol Logout yang aman.

Admin_Dashboard.cs — Navigasi Sidebar Admin

```
class="tok-kw">private class="tok-kw">void btnAkun_Click(object sender, EventArgs e)
{
    class="tok-kw">new Admin_Akun().Show();
    this.Close();
}

class="tok-kw">private class="tok-kw">void btnProduk_Click(object sender, EventArgs e)
{
    class="tok-kw">new Admin_Produk().Show();
    this.Close();
}

class="tok-kw">private class="tok-kw">void btnProfil_Click(object sender, EventArgs e)
{
    class="tok-kw">new Admin_Profil().Show();
    this.Close();
}

class="tok-kw">private class="tok-kw">void btnLogout_Click(object sender, EventArgs e)
{
    UserSession.LakukanLogout(this);
}
```

Admin_Akun.cs & Admin_FormAkun.cs (Admin_Akun.cs / Admin_FormAkun.cs)

MuatDataAkun()

METHOD (TABEL AKUN)

BUAT APA FUNGSI INI?

Menampilkan seluruh akun staf dari tabel Users ke DataGridView, lengkap dengan filter pencarian nama dan filter role (Admin/Kasir/Owner).

Admin_Akun.cs / Admin_FormAkun.cs — MuatDataAkun()

```
class="tok-kw">private class="tok-kw">void MuatDataAkun()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string query = class="tok-class="tok-kw">string">@"
            SELECT Id, Nama, Role
            FROM Users
            WHERE (@nama = '' OR Nama LIKE @keyword)
              AND (@role = 'Semua Role' OR Role = @role)
            ORDER BY Id DESC";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama",
txtCariNama.Text.Trim());
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@keyword", class="tok-
class="tok-kw">string">"%" + txtCariNama.Text.Trim() + class="tok-class="tok-kw">string">"%");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@role",
cmbCariRole.SelectedItem?.ToString() ?? class="tok-class="tok-kw">string">"Semua Role");

            class="tok-kw">using (SqlDataAdapter da = class="tok-kw">new SqlDataAdapter(cmd))
            {
                DataTable dt = class="tok-kw">new DataTable();
                da.Fill(dt);
                dgvDataProduk.DataSource = dt;
            }
        }
    }
}
```

Admin_FormAkun -> btnSimpan_Click(...)

EVENT HANDLER (POPUP SIMPAN AKUN)

BUAT APA FUNGSI INI?

Menyimpan akun baru atau mengupdate akun lama. Jika memilih foto profil, file gambar diubah menjadi biner byte[] untuk disimpan ke kolom FotoProfil (VARBINARY).

Admin_Akun.cs / Admin_FormAkun.cs — Admin_FormAkun -> btnSimpan_Click(...)

```
class="tok-kw">private class="tok-kw">void btnSimpan_Click(object sender, EventArgs e)
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">byte[]? fotoBytes = null;
        class="tok-kw">if (!class="tok-kw">string.IsNullOrEmpty(jalurFoto) &&
File.Exists(jalurFoto))
            fotoBytes = File.ReadAllBytes(jalurFoto);

        class="tok-kw">if (idEdit == null) class="tok-comment">// Tambah Akun Baru
        {
            class="tok-kw">string sql = class="tok-class="tok-kw">string">"INSERT INTO Users (Nama,
Password, Role, FotoProfil) VALUES (@nama, @pass, @role, @foto)";
            class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
            {
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama",
txtNama.Text.Trim());
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@pass",
textPass.Text.Trim());
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@role",
cmbRole.Text);
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@foto", fotoBytes !=
null ? (object)fotoBytes : DBNull.Value);
                cmd.ExecuteNonQuery();
            }
            Databases.CatatLog($class="tok-class="tok-kw">string">"Menambahkan akun {cmbRole.Text}:
{txtNama.Text.Trim()}");
        }
        class="tok-kw">else class="tok-comment">// Update Akun
        {
            class="tok-kw">string sql = class="tok-class="tok-kw">string">"UPDATE Users SET Nama =
@nama, Role = @role WHERE Id = @id";
            class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
            {
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@id", idEdit.Value);
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama",
txtNama.Text.Trim());
                cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@role",
cmbRole.Text);
                cmd.ExecuteNonQuery();
            }
            Databases.CatatLog($class="tok-class="tok-kw">string">"Memperbarui akun:
{txtNama.Text.Trim()}");
        }
```

```
        }
        this.DialogResult = DialogResult.OK;
        this.Close();
    }
```

Admin_Produk.cs & Admin_FormProduk.cs (Admin_Produk.cs / Admin_FormProduk.cs)

MuatDataProduk()

METHOD (TABEL OBAT)

BUAT APA FUNGSI INI?

Mengambil data seluruh obat. Kolom FotoProduk otomatis disetel sebagai thumbnail bergambar (Zoom), filter kategori obat, dan status stok (Tersedia, Menipis, Habis).

Admin_Produk.cs / Admin_FormProduk.cs — MuatDataProduk()

```
class="tok-kw">private class="tok-kw">void MuatDataProduk()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string query = class="tok-class="tok-kw">string">@"
            SELECT Id, FotoProduk, Nama, Category AS Kategori, TanggalKadaluarsa, Stok, Jenis,
Harga, Deskripsi
            FROM Produk
            WHERE (@nama = '' OR Nama LIKE @keyword)
              AND (@kategori = 'Semua Kategori' OR Category = @kategori)
            ORDER BY Id DESC";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama",
txtCariNama.Text.Trim());
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@keyword", class="tok-
class="tok-kw">string">"%" + txtCariNama.Text.Trim() + class="tok-class="tok-kw">string">"%");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@kategori",
cmbCariCategory.SelectedItem?.ToString() ?? class="tok-class="tok-kw">string">"Semua Kategori");

            DataTable dt = class="tok-kw">new DataTable();
            class="tok-kw">new SqlDataAdapter(cmd).Fill(dt);
            dgvDataProduk.DataSource = dt;

            class="tok-comment">// Atur gambar foto produk agar tampil cantik sebagai thumbnail
            class="tok-kw">if (dgvDataProduk.Columns[class="tok-class="tok-
kw">string">"FotoProduk"] is DataGridViewImageColumn imgCol)
            {
                imgCol.HeaderText = class="tok-class="tok-kw">string">"Foto";
                imgCol.ImageLayout = DataGridViewImageCellLayout.Zoom;
                imgCol.Width = 70;
                imgCol.DisplayIndex = 0;
            }
            dgvDataProduk.RowTemplate.Height = 60;
        }
    }
}
```

Admin_FormProduk -> btnSimpan_Click(...)

EVENT HANDLER (POPUP SIMPAN OBAT)

BUAT APA FUNGSI INI?

Menyimpan data obat baru atau mengupdate obat yang diedit, termasuk upload gambar kemasan obat ke format biner SQL Server.

Admin_Produk.cs / Admin_FormProduk.cs — Admin_FormProduk -> btnSimpan_Click(...)

```
class="tok-kw">private class="tok-kw">void btnSimpan_Click(object sender, EventArgs e)
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">byte[]? fotoBytes = null;
        class="tok-kw">if (!class="tok-kw">string.IsNullOrEmpty(jalurFoto) &&
File.Exists(jalurFoto))
            fotoBytes = File.ReadAllBytes(jalurFoto);

        class="tok-kw">string sql = class="tok-class="tok-kw">string">@"
            INSERT INTO Produk (IdAdmin, Nama, Category, TanggalKadaluarsa, Stok, Jenis, Harga,
Deskripsi, FotoProduk)
            VALUES (@admin, @nama, @cat, @exp, @stok, @jenis, @harga, @desk, @foto)";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@admin", UserSession.Id
> 0 ? UserSession.Id : 1);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama",
txtNama.Text.Trim());
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@cat",
cmbCategory.Text);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@exp",
dtpKadaluarsa.Value.Date);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@stok", class="tok-
kw">int.Parse(txtStok.Text.Trim()));
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@jenis", class="tok-
class="tok-kw">string">"Obat");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@harga",
txtHarga.Text.Trim());
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@desk", class="tok-
class="tok-kw">string">"Obat Apotek");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@foto", fotoBytes !=
null ? (object)fotoBytes : DBNull.Value);
            cmd.ExecuteNonQuery();
        }
        Databases.CatatLog($class="tok-class="tok-kw">string">"Menambahkan obat baru:
{txtNama.Text.Trim()}");
    }
    this.DialogResult = DialogResult.OK;
    this.Close();
}
```

Admin_Profil.cs (Admin_Profil.cs)

MuatDataProfil() & btnSimpan_Click(...)

METHOD & EVENT

BUAT APA FUNGSI INI?

MuatDataProfil menampilkan nama, role, dan foto profil admin. Tombol simpan memproses pembaruan password baru jika admin ingin mengganti kata sandi.

Admin_Profil.cs — MuatDataProfil() & btnSimpan_Click(...)

```
class="tok-kw">private class="tok-kw">void MuatDataProfil()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string sql = class="tok-class="tok-kw">string">"SELECT TOP 1 Id, Nama, Role,
FotoProfil FROM Users WHERE Id = @id";
        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@id", UserSession.Id);
            class="tok-kw">using (SqlDataReader r = cmd.ExecuteReader())
            {
                class="tok-kw">if (r.Read())
                {
                    lblNama.Text = r[class="tok-class="tok-kw">string">"Nama"].ToString();
                    lblRole.Text = r[class="tok-class="tok-kw">string">"Role"].ToString();
                    class="tok-kw">if (r[class="tok-class="tok-kw">string">"FotoProfil"] !=
DBNull.Value)
                    {
                        class="tok-kw">byte[] bytes = (class="tok-kw">byte[])r[class="tok-
class="tok-kw">string">"FotoProfil"];
                        class="tok-kw">using (MemoryStream ms = class="tok-kw">new
MemoryStream(bytes))
                            fotoProfil.Image = class="tok-kw">new Bitmap(Image.FromStream(ms));
                    }
                }
            }
        }
    }
}
```

MODUL KASIR

Bab 3: Modul Kasir (Point of Sale & Transaksi Penjualan)

Jantung operasional penjualan obat: pencatatan keranjang belanja sementara, cek ketersediaan stok fisik realtime, transaksi database ACID, cetak nota struk, dan pendaftaran pelanggan baru.

Kasir_Transaksi.cs (Kasir_Transaksi.cs)

`InisialisasiTabelKeranjang()` METHOD

BUAT APA FUNGSI INI?

Membuat tabel sementara (DataTable di memori) dengan kolom No, Nama Obat, Harga Satuan, Jumlah, dan Subtotal untuk ditampilkan di DataGridView belanjaan.

Kasir_Transaksi.cs — InisialisasiTabelKeranjang()

```
class="tok-kw">private class="tok-kw">void InisialisasiTabelKeranjang()
{
    dtKeranjang = class="tok-kw">new DataTable();
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"IdProduk", typeof(class="tok-
kw">int));
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"No", typeof(class="tok-kw">int));
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"Nama Obat", typeof(class="tok-
kw">string));
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"Harga Satuan", typeof(class="tok-
kw">int));
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"Jumlah", typeof(class="tok-
kw">int));
    dtKeranjang.Columns.Add(class="tok-class="tok-kw">string">"Subtotal", typeof(class="tok-
kw">int));

    dataGridView1.DataSource = dtKeranjang;
    dataGridView1.SelectionMode = DataGridViewSelectionMode.FullRowSelect;
    dataGridView1.ReadOnly = true;
    dataGridView1.AllowUserToAddRows = false;
    dataGridView1.Columns[class="tok-class="tok-kw">string">"IdProduk"].Visible = false;
class="tok-comment">// Sembunyikan ID teknis
}
```

`btnCatat_Click(object sender, EventArgs e)`

EVENT HANDLER (TOMBOL CATAT OBAT)

BUAT APA FUNGSI INI?

Mencatat obat yang dipilih ke keranjang belanja. Mengecek stok di database agar kasir tidak menjual melebihi stok yang ada di apotek, lalu menghitung subtotal.

Kasir_Transaksi.cs — btnCatat_Click(object sender, EventArgs e)

```
class="tok-kw">private class="tok-kw">void btnCatat_Click(object sender, EventArgs e)
{
    class="tok-kw">string namaBarang = txtBarang.Text.Trim();
    class="tok-kw">if (!class="tok-kw">int.TryParse(txtJumlah.Text.Trim(), out class="tok-kw">int
qty) || qty <= 0)
    {
        MessageBox.Show(class="tok-class="tok-kw">string">"Masukkan jumlah obat yang valid!");
        class="tok-kw">return;
    }

    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string query = class="tok-class="tok-kw">string">"SELECT TOP 1 Id, Nama,
Harga, Stok FROM Produk WHERE Nama = @nama";
        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@nama", namaBarang);
            class="tok-kw">using (SqlDataReader reader = cmd.ExecuteReader())
            {
                class="tok-kw">if (!reader.Read())
                {
                    MessageBox.Show(class="tok-class="tok-kw">string">"Obat tidak ditemukan di
apotek!");
                    class="tok-kw">return;
                }

                class="tok-kw">int idProduk = Convert.ToInt32(reader[class="tok-class="tok-
kw">string">"Id"]);
                class="tok-kw">int stokTersedia = Convert.ToInt32(reader[class="tok-class="tok-
kw">string">"Stok"]);
                class="tok-kw">int hargaSatuan = BersihkanHarga(reader[class="tok-class="tok-
kw">string">"Harga"].ToString());

                class="tok-comment">// Cek akumulasi jumlah jika obat sudah ada di keranjang
                class="tok-kw">int qtyDiKeranjang = 0;
                class="tok-kw">foreach (DataRow r class="tok-kw">in dtKeranjang.Rows)
                    class="tok-kw">if ((class="tok-kw">int)r[class="tok-class="tok-
kw">string">"IdProduk"] == idProduk) qtyDiKeranjang += (class="tok-kw">int)r[class="tok-class="tok-
kw">string">"Jumlah"];

                class="tok-kw">if (qtyDiKeranjang + qty > stokTersedia)
                {
                    MessageBox.Show($class="tok-class="tok-kw">string">"Stok tidak cukup! Sisa
stok: {stokTersedia}, sudah di keranjang: {qtyDiKeranjang}");
```

```
                    class="tok-kw">return;
                }

                class="tok-comment">// Tambahkan baris baru ke tabel keranjang
                dtKeranjang.Rows.Add(idProduk, dtKeranjang.Rows.Count + 1, namaBarang, hargaSatuan,
qty, qty * hargaSatuan);
                SusunUlangNomor();
                HitungTotal();

                txtBarang.Clear();
                txtJumlah.Text = class="tok-class="tok-kw">string">"1";
            }
        }
    }
}
```

HitungTotal() & HitungKembalian()

METHOD

BUAT APA FUNGSI INI?

Menjumlahkan seluruh subtotal di keranjang, lalu menghitung uang kembalian secara otomatis saat kasir mengetik uang tunai yang diserahkan pelanggan.

Kasir_Transaksi.cs — HitungTotal() & HitungKembalian()

```
class="tok-kw">private class="tok-kw">void HitungTotal()
{
    grandTotal = 0;
    class="tok-kw">foreach (DataRow row class="tok-kw">in dtKeranjang.Rows)
        grandTotal += Convert.ToInt32(row[class="tok-class="tok-kw">string">"Subtotal"]);

    lblHargaTotal.Text = class="tok-class="tok-kw">string">"Rp " + grandTotal.ToString(class="tok-
class="tok-kw">string">"N0", idCulture);
    HitungKembalian();
}

class="tok-kw">private class="tok-kw">void HitungKembalian()
{
    class="tok-kw">string raw = txtUangYangDibayar.Text.Replace(class="tok-class="tok-
kw">string">".", class="tok-class="tok-kw">string">"").Replace(class="tok-class="tok-
kw">string">"Rp", class="tok-class="tok-kw">string">"").Trim();
    class="tok-kw">if (class="tok-kw">int.TryParse(raw, out class="tok-kw">int uangBayar))
    {
        class="tok-kw">int kembalian = uangBayar - grandTotal;
        lblKembalian.ForeColor = kembalian >= 0 ? Color.Black : Color.Red;
        lblKembalian.Text = class="tok-class="tok-kw">string">"Rp " +
kembalian.ToString(class="tok-class="tok-kw">string">"N0", idCulture);
    }
}
```

btnCetakStruk_Click(...) (Simpan & Cetak)

EVENT HANDLER (TOMBOL CETAK TRANSAKSI)

BUAT APA FUNGSI INI?

Puncak transaksi: Menggunakan SqlTransaction untuk menyimpan data ke Transaksi, menyimpan tiap item ke DetailTransaksi, memotong stok Produk, menambah poin pelanggan, mencatat log, lalu menampilkan pop-up struk.

Kasir_Transaksi.cs — btnCetakStruk_Click(...) (Simpan & Cetak)

```
class="tok-kw">private class="tok-kw">void btnCetakStruk_Click(object sender, EventArgs e)
{
    class="tok-kw">if (dtKeranjang.Rows.Count == 0) class="tok-kw">return;
    class="tok-kw">int uangBayar = class="tok-
kw">int.Parse(txtUangYangDibayar.Text.Replace(class="tok-class="tok-kw">string">".", class="tok-
class="tok-kw">string">""));
    class="tok-kw">int kembalian = uangBayar - grandTotal;

    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">using (SqlTransaction trans = conn.BeginTransaction())
        {
            class="tok-kw">try
            {
                class="tok-comment">// 1. Simpan Transaksi Utama
                class="tok-kw">string qTrans = class="tok-class="tok-kw">string">@"
                    INSERT INTO Transaksi (IdKasir, IdPelanggan, TotalProduk, Harga_Total,
Uang_Bayar, Uang_Kembali, Tanggal_Transaksi)
                    OUTPUT INSERTED.Id
                    VALUES (@idKasir, @idPelanggan, @totalQty, @totalHarga, @bayar, @kembali,
GETDATE())";

                class="tok-kw">int idTransaksiBaru;
                class="tok-kw">using (SqlCommand cmdT = class="tok-kw">new SqlCommand(qTrans, conn,
trans))
                {
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@idKasir",
UserSession.Id);
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@idPelanggan",
idPelangganVal);
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@totalQty",
totalQty);
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@totalHarga",
grandTotal);
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@bayar",
uangBayar);
                    cmdT.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@kembali",
kembalian);
                    idTransaksiBaru = (class="tok-kw">int)cmdT.ExecuteScalar();
                }

                class="tok-comment">// 2. Simpan DetailTransaksi & Potong Stok Produk
                class="tok-kw">foreach (DataRow r class="tok-kw">in dtKeranjang.Rows)
```

```
                {
                    int idProduk = (int)r["IdProduk"];
                    int qty = (int)r["Jumlah"];
                    int harga = (int)r["Harga Satuan"];

                    string qDetail = "INSERT INTO DetailTransaksi (IdTransaksi, IdProduk, HargaSatuan, Jumlah) VALUES (@t, @p, @h, @j)";
                    using (SqlCommand cmdD = new SqlCommand(qDetail, conn, trans))
                    {
                        cmdD.Parameters.AddWithValue("@t", idTransaksiBaru);
                        cmdD.Parameters.AddWithValue("@p", idProduk);
                        cmdD.Parameters.AddWithValue("@h", harga);
                        cmdD.Parameters.AddWithValue("@j", qty);
                        cmdD.ExecuteNonQuery();
                    }

                    // Potong stok fisik obat
                    string qStok = "UPDATE Produk SET Stok = Stok - @qty WHERE Id = @p";
                    using (SqlCommand cmdS = new SqlCommand(qStok, conn, trans))
                    {
                        cmdS.Parameters.AddWithValue("@qty", qty);
                        cmdS.Parameters.AddWithValue("@p", idProduk);
                        cmdS.ExecuteNonQuery();
                    }
                }

                trans.Commit(); // Berhasil simpan semua data secara utuh!

                // 3. Tampilkan Pop-up Struk Kasir
                string struk = BuatTeksStruk(idTransaksiBaru, txtPelanggan.Text, uangBayar, kembalian);
                new Kasir_StrukDialog(struk).ShowDialog();

                dtKeranjang.Clear();
                grandTotal = 0;
            }
            catch (Exception ex)
            {
                trans.Rollback();
                MessageBox.Show("Gagal menyimpan transaksi: " + ex.Message);
            }
        }
```

```
    }
}
```

Kasir_FormPelanggan.cs (Pop-up Registrasi) (Kasir_FormPelanggan.cs)

btnSimpan_Click(object sender, EventArgs e) EVENT HANDLER (SIMPAN PELANGGAN BARU)

BUAT APA FUNGSI INI?
Mendaftarkan pelanggan baru jika belum terdaftar di apotek, memvalidasi nama, email, dan nomor HP agar tidak duplikat, lalu otomatis memilih pelanggan tersebut di kasir.

Kasir_FormPelanggan.cs — btnSimpan_Click(object sender, EventArgs e)

```
class="tok-kw">private class="tok-kw">void btnSimpan_Click(object sender, EventArgs e)
{
    class="tok-kw">string nama = txtNama.Text.Trim();
    class="tok-kw">string email = txtEmail.Text.Trim();
    class="tok-kw">string noHp = txtNomorHP.Text.Trim();

    class="tok-kw">if (class="tok-kw">string.IsNullOrWhiteSpace(nama) || class="tok-
kw">string.IsNullOrWhiteSpace(email) || class="tok-kw">string.IsNullOrWhiteSpace(noHp))
    {
        MessageBox.Show(class="tok-class="tok-kw">string">"Nama, Email, dan No HP wajib diisi!");
        class="tok-kw">return;
    }

    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string sql = class="tok-class="tok-kw">string">"INSERT INTO Pelanggan
(IdKasir, Nama, Email, NomorHP, TotalTransaksi) VALUES (@k, @n, @e, @hp, 0)";
        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@k", UserSession.Id);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@n", nama);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@e", email);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@hp", noHp);
            cmd.ExecuteNonQuery();
        }
        Databases.CatatLog($class="tok-class="tok-kw">string">"Mendaftarkan pelanggan baru:
{nama}");
    }

    NamaPelangganBaru = nama;
    this.DialogResult = DialogResult.OK;
    this.Close();
}
```

Kasir_StrukDialog.cs (Pop-up Struk Kasir) (Kasir_StrukDialog.cs)

AturTengahStruk() & btnCetak_Click(...)

METHOD & EVENT

BUAT APA FUNGSI INI?

AturTengahStruk menghitung indentasi margin kiri dan kanan agar struk belanja tampil persis di tengah jendela. Tombol cetak mengirimkan nota ke printer fisik atau PDF.

Kasir_StrukDialog.cs — AturTengahStruk() & btnCetak_Click(...)

```
class="tok-kw">private class="tok-kw">void AturTengahStruk()
{
    class="tok-kw">using (Graphics g = rtbStruk.CreateGraphics())
    {
        SizeF sz = g.MeasureString(class="tok-class="tok-
kw">string">"========================================", rtbStruk.Font);
        class="tok-kw">int sisaLebar = rtbStruk.ClientSize.Width - (class="tok-kw">int)sz.Width;
        class="tok-kw">int indent = Math.Max(15, sisaLebar / 2);

        rtbStruk.SelectAll();
        rtbStruk.SelectionIndent = indent;
        rtbStruk.SelectionRightIndent = indent;
        rtbStruk.DeselectAll();
    }
}

class="tok-kw">private class="tok-kw">void btnCetak_Click(object sender, EventArgs e)
{
    PrintDocument pd = class="tok-kw">new PrintDocument();
    pd.PrintPage += (s, ev) => {
        class="tok-kw">using (Font font = class="tok-kw">new Font(class="tok-class="tok-
kw">string">"Courier New", 9))
        {
            SizeF sz = ev.Graphics.MeasureString(class="tok-class="tok-
kw">string">"========================================", font);
            float x = Math.Max(10, (ev.PageBounds.Width - sz.Width) / 2);
            ev.Graphics.DrawString(teksStruk, font, Brushes.Black, class="tok-kw">new PointF(x,
20));
        }
    };
    class="tok-kw">new PrintDialog { Document = pd }.ShowDialog();
}
```

MODUL OWNER

Bab 4: Modul Owner (Laporan Keuangan & Audit Aktivitas)

Modul pemilik apotek untuk memantau performa penjualan, melihat laporan transaksi terinci, mengawasi status stok obat, dan memeriksa audit trail aktivitas seluruh pegawai.

Owner_Dashboard.cs (Owner_Dashboard.cs)

LoadStatistik() & LoadAktivitas()

METHOD

BUAT APA FUNGSI INI?

Menampilkan statistik omzet & inventori toko, serta tabel 10 aktivitas terkini pegawai apotek menggunakan LEFT JOIN Users agar log tetap utuh.

Owner_Dashboard.cs — LoadStatistik() & LoadAktivitas()

```
class="tok-kw">private class="tok-kw">void LoadAktivitas()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string sql = class="tok-class="tok-kw">string">@"
            SELECT TOP 10
                L.Id AS [ID Log],
                COALESCE(U.Nama, 'Sistem') AS [Pengguna],
                L.Aktivitas AS [Aktivitas],
                L.Waktu AS [Waktu]
            FROM LogAktivitas L
            LEFT JOIN Users U ON L.IdUser = U.Id
            ORDER BY L.Waktu DESC";

        DataTable dt = class="tok-kw">new DataTable();
        class="tok-kw">new SqlDataAdapter(sql, conn).Fill(dt);
        dgv10AktivitasTerkini.DataSource = dt;
    }
}
```

Owner_Transaksi.cs (Owner_Transaksi.cs)

`TampilkanTransaksi()`

METHOD (LAPORAN TRANSAKSI)

BUAT APA FUNGSI INI?

Menampilkan riwayat transaksi lengkap dengan menggabungkan nama-nama obat per transaksi menggunakan STRING_AGG SQL Server, serta filter rentang tanggal, pelanggan, dan obat.

Owner_Transaksi.cs — TampilkanTransaksi()

```
class="tok-kw">private class="tok-kw">void TampilkanTransaksi()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string query = class="tok-class="tok-kw">string">@"
            SELECT
                t.Id AS 'ID Transaksi',
                COALESCE(pl.Nama, 'Umum') AS 'Pelanggan',
                COALESCE(
                    (SELECT STRING_AGG(p2.Nama, ', ')
                     FROM DetailTransaksi dt2
                     INNER JOIN Produk p2 ON dt2.IdProduk = p2.Id
                     WHERE dt2.IdTransaksi = t.Id), '-'
                ) AS 'Produk',
                t.TotalProduk AS 'Total Item',
                t.Harga_Total AS 'Total Harga',
                t.Uang_Bayar AS 'Uang Bayar',
                t.Uang_Kembali AS 'Uang Kembali',
                t.Tanggal_Transaksi AS 'Tanggal Transaksi',
                u.Nama AS 'Kasir'
            FROM Transaksi t
            LEFT JOIN Pelanggan pl ON t.IdPelanggan = pl.Id
            LEFT JOIN Users u ON t.IdKasir = u.Id
            WHERE t.Tanggal_Transaksi >= @tglAwal AND t.Tanggal_Transaksi < @tglAkhir
            ORDER BY t.Tanggal_Transaksi DESC";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@tglAwal",
dtpFilterAwal.Value.Date);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@tglAkhir",
dtpFilterAkhir.Value.Date.AddDays(1));

            DataTable dt = class="tok-kw">new DataTable();
            class="tok-kw">new SqlDataAdapter(cmd).Fill(dt);
            dgvDataProduk.DataSource = dt;
        }
    }
}
```

Owner_LogAktivitas.cs (Owner_LogAktivitas.cs)

LoadDataLog()

METHOD (AUDIT TRAIL LOG)

BUAT APA FUNGSI INI?

Menampilkan seluruh rekam jejak aktivitas operasional toko dengan filter kategori (Semua, Login/Logout, Transaksi, Kelola Produk, Kelola Akun, Pelanggan) dan rentang tanggal.

Owner_LogAktivitas.cs — LoadDataLog()

```
class="tok-kw">private class="tok-kw">void LoadDataLog()
{
    class="tok-kw">using (SqlConnection conn = Databases.GetConn())
    {
        conn.Open();
        class="tok-kw">string sql = class="tok-class="tok-kw">string">@"
            SELECT
                L.Id AS [ID Log],
                COALESCE(U.Nama, 'Sistem') AS [Nama Pegawai],
                L.Aktivitas AS [Aktivitas],
                L.Waktu AS [Waktu]
            FROM LogAktivitas L
            LEFT JOIN Users U ON U.Id = L.IdUser
            WHERE
                (@cari = '' OR U.Nama LIKE @keyword OR L.Aktivitas LIKE @keyword)
                AND (
                    @kategori = 'Semua Aktivitas'
                    OR (@kategori = 'Login & Logout' AND (L.Aktivitas LIKE '%Login%' OR L.Aktivitas
LIKE '%Logout%'))
                    OR (@kategori = 'Transaksi' AND L.Aktivitas LIKE '%Transaksi%')
                    OR (@kategori = 'Kelola Produk' AND (L.Aktivitas LIKE '%obat%' OR L.Aktivitas
LIKE '%Produk%'))
                    OR (@kategori = 'Kelola Akun' AND (L.Aktivitas LIKE '%akun%' OR L.Aktivitas
LIKE '%profil%'))
                    OR (@kategori = 'Pelanggan' AND L.Aktivitas LIKE '%pelanggan%')
                )
                AND L.Waktu >= @awal AND L.Waktu < @akhir
            ORDER BY L.Waktu DESC";

        class="tok-kw">using (SqlCommand cmd = class="tok-kw">new SqlCommand(sql, conn))
        {
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@cari",
txtCariNama.Text.Trim());
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@keyword", class="tok-
class="tok-kw">string">"%" + txtCariNama.Text.Trim() + class="tok-class="tok-kw">string">"%");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@kategori",
cmbCariCategory.SelectedItem?.ToString() ?? class="tok-class="tok-kw">string">"Semua Aktivitas");
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@awal",
dtpFilterAwal.Value.Date);
            cmd.Parameters.AddWithValue(class="tok-class="tok-kw">string">"@akhir",
dtpFilterAkhir.Value.Date.AddDays(1));

            DataTable dt = class="tok-kw">new DataTable();
            class="tok-kw">new SqlDataAdapter(cmd).Fill(dt);
            dgvDataProduk.DataSource = dt;
```

```
        }
    }
}
```